

Python Step- by- Step Course (Hands- on)

Lesson 1 — Getting Started & Variables

You'll learn: running Python, print, variables, basic types, operators, input, f- strings, comments, common errors.

Your first script: hello.py

```
print("Hello, Python!")
```

Run it. If you see the message, you've begun ✓

Variables & Basic Types

- Python has **dynamic typing**. Examples:

```
age = 20          # int
height = 1.75     # float
name = "Asha"     # str
is_student = True # bool
```

- Check types:

```
print(type(age))
```

Operators

+ - * / // % **

```
print(7 / 2)    # 3.5 (float division)
print(7 // 2)   # 3   (floor division)
print(7 % 2)    # 1   (remainder)
print(2 ** 3)   # 8   (power)
```

Input & Casting

```
name = input("Your name: ")      # returns str
year = int(input("Your birth year: ")) # cast to int
print(f"Hi {name}, you are {2025 - year} years old (approx).")
```

Comments & Style

- Single- line comment: # this is a comment

Common gotchas

- input() returns **str** → cast with int() / float() for math.
 - ZeroDivisionError, NameError (misspelled variable), IndentationError.
-

Practice — Lesson 1

Try these in separate files or one by one. **Do not scroll to solutions until you try!**

- 1) **Personal greeting**: Ask the user's name and print Hello, <name>!.
- 2) **Rectangle**: Read length & width; print **area** and **perimeter**.
- 3) **Celsius → Fahrenheit**: $F = C * 9/5 + 32$.
- 4) **Simple Interest**: Read P, R, T; compute $SI = P * R * T / 100$ and Amount = P + SI.
- 5) **Seconds → H:MM:SS**: Convert a total number of seconds to hours, minutes, seconds with zero- padded minutes/seconds.
- 6) **Swap**: Read two values a and b, swap them **without** a temp variable, then print.
- 7) **Sum of digits**: For a 3- digit number, print the sum of its digits.
- 8) **Age in days (approx)**: Read age in years, output days (use 365).
- 9) **Tip calculator**: Given bill amount and tip %, output tip and total.
- 10) **Operator precedence**: Compute and print $2 + 3 * 4$, $(2 + 3) * 4$, and $2 ** 3 ** 2$.

Solutions — Lesson 1 (peek only after attempting)

1) Personal greeting

```
name = input("Your name: ")
print(f"Hello, {name}!")
```

2) Rectangle

```
length = float(input("Length: "))
width = float(input("Width: "))
area = length * width
perimeter = 2 * (length + width)
print(f"Area = {area}, Perimeter = {perimeter}")
```

3) Celsius → Fahrenheit

```
c = float(input("Celsius: "))
f = c * 9/5 + 32
print(f"{c}°C is {f}°F")
```

4) Simple Interest

```
p = float(input("Principal: "))
r = float(input("Rate (%): "))
```

```
t = float(input("Time (years): "))
si = p * r * t / 100
amount = p + si
print(f"Simple Interest = {si}")
print(f"Amount = {amount}")
```

5) Seconds → H:MM:SS

```
total = int(input("Total seconds: "))
h = total // 3600
m = (total % 3600) // 60
s = total % 60
print(f"{h:02d}:{m:02d}:{s:02d}")
```

6) Swap values

```
a = input("a: ")
b = input("b: ")
a, b = b, a
print("a =", a, "b =", b)
```

7) Sum of digits (3- digit)

```
n = int(input("3-digit number: "))
hundreds = n // 100
tens = (n // 10) % 10
ones = n % 10
print(hundreds + tens + ones)
```

8) Age in days

```
years = int(input("Age in years: "))
days = years * 365
print(days)
```

9) Tip calculator

```
bill = float(input("Bill amount: "))
tip_percent = float(input("Tip %: "))
tip = bill * tip_percent / 100
total = bill + tip
print(f"Tip = {tip}")
print(f"Total = {total}")
```

10) Precedence

```
a = 2 + 3 * 4
b = (2 + 3) * 4
c = 2 ** 3 ** 2 # right-associative: 2 ** (3 ** 2) = 2 ** 9 = 512
print(a, b, c) # 14 20 512
```

Mini- Project — Unit Converter (CLI)

Goal: A simple menu- driven converter for: km↔mi, kg↔lb, °C↔°F.

Spec - Show a numbered menu - Ask for a choice and a value - Convert and print the result - Repeat until the user chooses Exit

Starter

```
while True:
    print("\nUnit Converter")
    print("1) km → miles\n2) miles → km\n3) kg → lb\n4) lb → kg\n5) °C → °F\n6) °F → °C\n0) Exit")
    choice = input("Choose: ")

    if choice == "0":
        print("Goodbye!")
        break

    value = float(input("Enter value: "))

    if choice == "1":
        print(value * 0.621371)
    elif choice == "2":
        print(value / 0.621371)
    elif choice == "3":
        print(value * 2.20462)
    elif choice == "4":
        print(value / 2.20462)
    elif choice == "5":
        print(value * 9/5 + 32)
    elif choice == "6":
        print((value - 32) * 5/9)
    else:
        print("Invalid choice.")
```

Quick Quiz — Lesson 1

1. What type does `input()` return by default?
2. What's the difference between `/` and `//`?
3. Which operator is used for exponentiation?
4. Which is a valid identifier: `2cool`, `cool_2`, `cool-2`?
5. How do you write a single- line comment?

Answers 1) str 2) / true division (float), // floor division 3) **
4) cool_2 5) With #